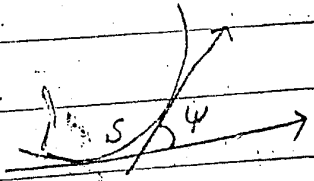


7-7-1

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www.mathcity.orgCurvature and Radius of Curvature 7.7

Rate of change of ψ with respect to s is called curvature, K .

$$K = \left| \frac{d\psi}{ds} \right|$$



s is the arc length

ψ is the inclination of the tangent to the curve.

The reciprocal of curvature is called Radius of Curvature denoted by ρ

$$K = \frac{1}{\rho} \quad \text{or} \quad \rho = \frac{1}{K}$$

1) When the eq. of the curve is given in cartesian form

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}{\frac{d^2y}{dx^2}}$$

$$K = \frac{\frac{d^2y}{dx^2}}{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}$$

2) When the curve is given in implicit form

$$\rho = \frac{(f_x^2 + f_y^2)^{3/2}}{|(f_y)(f_{xx}) - 2f_x f_y f_{xy} + (f_x^2) f_{yy}|}$$

$$K = \frac{1}{\rho}$$

3) When the curve is given in Parametric form.

$$\rho = \frac{[(f')^2 + (g')^2]^{3/2}}{|f'g'' - g'f''|}$$

$$\rho = \frac{\left[\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right]^{3/2}}{\left| \frac{dx}{dt} \cdot \frac{d^2y}{dt^2} - \frac{dy}{dt} \cdot \frac{d^2x}{dt^2} \right|}$$

$$K = \frac{|f'g'' - g'f''|}{[(f')^2 + (g')^2]^{3/2}}$$

7.7-2

When the eq of curve is given in Polar form.

$$\rho = \frac{(r^2 + (r')^2)^{3/2}}{(r^2 + 2(r'r'') - r r''')}$$

$$r' = \frac{dr}{d\theta}$$

$$r'' = \frac{d^2r}{d\theta^2}$$

$$K = \frac{1}{\rho}$$

Also,

$$\rho = r \frac{dr}{dp} \quad \text{where } p = r \sin \phi$$

$$K = \frac{1}{r} \frac{dp}{dr}$$

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Ex 7.7

① $y = c \cosh x$

$$\rho = \frac{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}{\frac{d^2y}{dx^2}}$$

$$\frac{dy}{dx} = \frac{1}{c} \sinh x$$

$$\frac{d^2y}{dx^2} = \frac{1}{c} \cosh x$$

$$= \left(1 + \sinh^2 \frac{x}{c}\right)^{3/2}$$

$$\frac{1}{c} \cosh \frac{x}{c}$$

$$\because \cosh^2 \frac{x}{c} - \sinh^2 \frac{x}{c} = 1$$

$$= \frac{\left(\cosh \frac{x}{c}\right)^{3/2}}{\frac{1}{c} \cosh \frac{x}{c}}$$

$$= \frac{\left(\cosh \frac{x}{c}\right)^3}{\frac{1}{c} \cosh \frac{x}{c}} = c \left(\cosh \frac{x}{c}\right)^2$$

7.7-3

but $\frac{y}{c} = \cosh \frac{x}{c}$

$\therefore \rho = c \frac{y^2}{c^2} = \frac{y^2}{c}$ Ans.

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② $x = a(\cos t + t \sin t)$ $y = a(\sin t - t \cos t)$

$\rho = \left[\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 \right]^{3/2}$

$\left| \frac{dx}{dt} \frac{d^2y}{dt^2} - \frac{dy}{dt} \frac{d^2x}{dt^2} \right|$

$= \left[(at \cos t)^2 + (at \sin t)^2 \right]^{3/2}$

$\left| (at \cos t) a(\sin t + t \cos t) - (at \sin t) a(\cos t - t \sin t) \right|$

$= a(\cos t - t \sin t)$

$\frac{dx}{dt} = a(-\sin t + \sin t + t \cos t)$

$= at \cos t$

$\frac{d^2x}{dt^2} = a \cos t + a t(-\sin t)$

$= a(\cos t - t \sin t)$

$\frac{dy}{dt} = a(\cos t - \cos t - t(-\sin t))$

$= at \sin t$

$\frac{d^2y}{dt^2} = a \sin t + at \cos t$

$= a(\sin t + t \cos t)$

$\rho = \left[a^2 t^2 \cos^2 t + a^2 t^2 \sin^2 t \right]^{3/2}$

$\left| a^2 t \cos t \sin t + a^2 t^2 \cos^2 t - a^2 t \sin t \cos t - a^2 t^2 \sin^2 t \right|$

$= \left[a^2 t^2 (\cos^2 t + \sin^2 t) \right]^{3/2}$

$\left| a^2 t^2 (\cos^2 t + \sin^2 t) \right|$

$= \left[(at)^2 \right]^{3/2} = (at)^3 = at$ Ans.

$\left| (at)^2 \right| \quad \left| (at)^3 \right|$

7.7-4

$$x = a \cos^3 \theta$$

$$y = a \sin^3 \theta$$

$$S = \left[\left(\frac{dx}{d\theta} \right)^2 + \left(\frac{dy}{d\theta} \right)^2 \right]^{3/2} \quad \text{--- ①}$$

$$\left| \frac{dx}{d\theta} \frac{d^2 y}{d\theta^2} - \frac{dy}{d\theta} \frac{d^2 x}{d\theta^2} \right|$$

$$\frac{dx}{d\theta} = 3a \cos^2 \theta (-\sin \theta)$$

$$\frac{d^2 x}{d\theta^2} = -3a [2 \cos \theta (-\sin \theta) (\sin \theta) + \cos^2 \theta \cos \theta]$$

$$= -3a \cos \theta [-2 \sin^2 \theta + \cos^2 \theta]$$

$$\frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$$

$$\frac{d^2 y}{d\theta^2} = 3a (2 \sin \theta \cos \theta \cos \theta + \sin^2 \theta (-\sin \theta))$$

$$= 3a \sin \theta (2 \cos^2 \theta - \sin^2 \theta)$$

Putting values in ①

$$S = \left[(3a \cos^2 \theta (-\sin \theta))^2 + (3a \sin^2 \theta \cos \theta)^2 \right]^{3/2}$$

$$\left| 3a \cos^2 \theta (-\sin \theta) 3a \sin \theta (2 \cos^2 \theta - \sin^2 \theta) - 3a \sin^2 \theta \cos \theta [-3a \cos \theta (-2 \sin^2 \theta + \cos^2 \theta)] \right|$$

$$= \left[9a^2 \cos^4 \theta \sin^2 \theta + 9a^2 \sin^4 \theta \cos^2 \theta \right]^{3/2}$$

$$\left| -9a^2 \sin^2 \theta \cos^2 \theta (2 \cos^2 \theta - \sin^2 \theta) + 9a^2 \sin^2 \theta \cos^2 \theta (-2 \sin^2 \theta + \cos^2 \theta) \right|$$

$$= \left[9a^2 \cos^2 \theta \sin^2 \theta (\cos^2 \theta + \sin^2 \theta) \right]^{3/2}$$

$$\left| 9a^2 \sin^2 \theta \cos^2 \theta [-2 \cos^2 \theta + \sin^2 \theta - 2 \sin^2 \theta + \cos^2 \theta] \right|$$

7.7-5

$$= \frac{(9a^2 \cos^2 \theta \sin^2 \theta)^{3/2}}{|9a^2 \sin^2 \theta \cos^2 \theta [-2(\cos^2 \theta + \sin^2 \theta) + (\sin^2 \theta + \cos^2 \theta)]|}$$

$$= \frac{(3a \cos \theta \sin \theta)^2}{|(3a \sin \theta \cos \theta)^2 [-2 + 1]|}$$

$$= \frac{(3a \cos \theta \sin \theta)^3}{-(3a \sin \theta \cos \theta)^2}$$

$$= -3a \cos \theta \sin \theta.$$

$$= 3a \left(\frac{x}{a}\right)^{1/3} \left(\frac{y}{a}\right)^{1/3}$$

$$= 3a \frac{x^{1/3}}{a^{1/3}} \frac{y^{1/3}}{a^{1/3}}$$

$$= 3a x^{1/3} y^{1/3} a^{2/3}$$

$$= 3a^{1-2/3} x^{1/3} y^{1/3}$$

$$= 3a^{1/3} x^{1/3} y^{1/3}$$

$$= 3(axy)^{1/3}$$

Parametric Form can be changed into Cartesian form by $\frac{dy}{dn} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dn}$.

and then using $s = \sqrt{1 + \left(\frac{dy}{dn}\right)^2}^{3/2}$
 $\frac{d^2y}{dn^2}$

7.7-6

$$r = 2 \cos 2\theta \quad \text{at } \theta = \frac{\pi}{12}$$

$$\frac{dr}{d\theta} = 2(-\sin 2\theta) \cdot 2 = -4 \sin 2\theta$$

$$\frac{d^2r}{d\theta^2} = -4 \cos 2\theta (2) = -8 \cos 2\theta$$

$$\rho = \left[r^2 + \left(\frac{dr}{d\theta} \right)^2 \right]^{3/2}$$

$$r^2 + 2 \left(\frac{dr}{d\theta} \right)^2 - r \frac{d^2r}{d\theta^2}$$

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$$\text{At } \theta = \frac{\pi}{12}$$

$$r = 2 \cos 2 \left(\frac{\pi}{12} \right) = 2 \cos \frac{\pi}{6} = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$$

$$\frac{dr}{d\theta} = -4 \sin 2 \left(\frac{\pi}{12} \right) = -4 \sin \frac{\pi}{6} = -4 \left(\frac{1}{2} \right) = -2$$

$$\frac{d^2r}{d\theta^2} = -8 \cos 2 \left(\frac{\pi}{12} \right) = -8 \cos \frac{\pi}{6} = -8 \frac{\sqrt{3}}{2} = -4\sqrt{3}$$

$$\rho = \left[3 + (-2)^2 \right]^{3/2}$$

$$3 + 2(-2)^2 - \sqrt{3}(-4\sqrt{3})$$

$$= (3+4)^{3/2}$$

$$= (7)^{3/2}$$

$$= \frac{17\sqrt{7}}{23} \text{ Ans.}$$

$$3+8+12$$

$$23$$

$$23$$

56 $r\theta = a$

$$-r = \frac{a}{\theta}$$

$$\frac{dr}{d\theta} = -\frac{a}{\theta^2}$$

$$\frac{d^2r}{d\theta^2} = +\frac{2a}{\theta^3}$$

$$\rho = \left(r^2 + \left(\frac{dr}{d\theta} \right)^2 \right)^{3/2}$$

$$r^2 + 2 \left(\frac{dr}{d\theta} \right)^2 = r \frac{d^2r}{d\theta^2}$$

$$\rho = \left(\frac{a^2}{\theta^2} + \left(-\frac{a}{\theta^2} \right)^2 \right)^{3/2}$$

$$\frac{a^2}{\theta^2} + 2 \left(\frac{a}{\theta^2} \right)^2 = \left(\frac{a}{\theta} \right) \left(\frac{2a}{\theta^3} \right)$$

$$\rho = \frac{\left(\frac{a^2}{\theta^2} + \frac{a^2}{\theta^4} \right)^{3/2}}{\frac{a^2}{\theta^2} + 2 \frac{a^2}{\theta^4} - \frac{2a^2}{\theta^4}} = \frac{\left(\frac{a^2 \theta^2 + a^2}{\theta^4} \right)^{3/2}}{\frac{a^2}{\theta^2}}$$

$$\rho = \frac{\left[a^2 (\theta^2 + 1) \right]^{3/2}}{(\theta^4)^{3/2}} = \frac{(a^2)^{3/2} (\theta^2 + 1)^{3/2}}{\theta^6} = \frac{a^2}{\theta^2}$$

$$= \frac{a^3 (\theta^2 + 1)^{3/2}}{\theta^6 a^2} = \frac{a (\theta^2 + 1)^{3/2}}{\theta^4} \text{ Ans.}$$

New Method Difficult

⑦ $r^n = a^n \sin n\theta$

$$n r^{n-1} \frac{dr}{d\theta} = a^n \cos n\theta (n)$$

$$r^n \cdot r' \cdot \frac{dr}{d\theta} = a^n \cos n\theta \frac{r^n}{n}$$

First find $\tan \psi = \frac{r}{\frac{dr}{d\theta}}$

then use $\rho = r \sin \psi$

then use $\rho = r \frac{dr}{dp}$

7.7-8

$$\frac{r^n}{r} \frac{dr}{d\theta} = a^n \cos n\theta.$$

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{a^n}{r^n} \cos n\theta.$$

$$r \frac{dr}{d\theta} = \frac{r^n}{a^n \cos n\theta}$$

$$r \frac{dr}{d\theta} = \frac{a^n \sin n\theta}{a^n \cos n\theta}$$

$$\therefore r = a^n \sin n\theta$$

$$\tan \phi = \tan n\theta.$$

$$\therefore \frac{r d\theta}{dr} = \tan \phi$$

$$\phi = n\theta$$

We know $p = r \sin \phi$

So $p = r \sin n\theta.$

$$\text{But } \sin(n\theta) = \frac{r^n}{a^n}$$

$$\therefore r^n = a^n \sin n\theta$$

$$\therefore p = r \frac{r^n}{a^n} = \frac{r^{n+1}}{a^n}$$

$$\frac{dp}{dr} = \frac{(n+1)r^n}{a^n}$$

$$\frac{dr}{dp} = \frac{a^n}{(n+1)r^n}$$

$$r \frac{dr}{dp} = \frac{r a^n}{(n+1)r^n}$$

$$p = \frac{a^n}{(n+1)r^{n-1}}$$

$$\therefore p = r \frac{dr}{dp}$$

7-7-9

other Method, Easy

⑦ $r^n = a^n \sin n\theta$

$$\rho = \frac{\left[r^2 + \left(\frac{dr}{d\theta} \right)^2 \right]^{3/2}}{r^2 + 2 \left(\frac{dr}{d\theta} \right)^2 - r \frac{d^2r}{d\theta^2}}$$

$$n r^{n-1} \frac{dr}{d\theta} = a^n \cos n\theta$$

$$\frac{dr}{d\theta} = \frac{a^n \cos n\theta}{r^{n-1}}$$

$$\frac{dr}{d\theta} = \frac{n a^n \cos n\theta}{r \cdot r^{n-1}} = \frac{n a^n \cos n\theta}{r^n \sin n\theta} \quad \because r = a^n \sin n\theta$$

$$\frac{dr}{d\theta} = r \cot n\theta$$

$$\frac{d^2r}{d\theta^2} = r (-\operatorname{cosec}^2 n\theta) + \frac{dr}{d\theta} \cot n\theta$$

$$= -n r \operatorname{cosec}^2 n\theta + (r \cot n\theta) (\cot n\theta)$$

$$\frac{d^2r}{d\theta^2} = r \cot^2 n\theta - n r \operatorname{cosec}^2 n\theta$$

$$\rho = \frac{\left[r^2 + r^2 \cot^2 n\theta \right]^{3/2}}{r^2 + 2(r^2 \cot^2 n\theta) - r \{ r \cot^2 n\theta - n r \operatorname{cosec}^2 n\theta \}}$$

$$= \frac{\left[r^2 (1 + \cot^2 n\theta) \right]^{3/2}}{r^2 + 2r^2 \cot^2 n\theta - r^2 \cot^2 n\theta + n r^2 \operatorname{cosec}^2 n\theta}$$

$$= r^3 (\operatorname{cosec}^2 n\theta)^{3/2}$$

$$r^2 + r^2 \cot^2 n\theta + n r^2 \operatorname{cosec}^2 n\theta$$

7.7-10

$$= \frac{r^3 \operatorname{Cosec}^3 n\theta}{r^2 \operatorname{Cosec}^2 n\theta + n r^2 \operatorname{Cosec}^2 n\theta}$$

$$= \frac{r^3 \operatorname{Cosec}^3 n\theta}{r^2 \operatorname{Cosec}^2 n\theta + n r^2 \operatorname{Cosec}^2 n\theta}$$

$$= \frac{r^3 \operatorname{Cosec}^3 n\theta}{r^2 \operatorname{Cosec}^2 n\theta + n r^2 \operatorname{Cosec}^2 n\theta}$$

$$= \frac{r^3 \operatorname{Cosec}^3 n\theta}{r^2 \operatorname{Cosec}^2 n\theta (1+n)}$$

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$$= \frac{r \operatorname{Cosec} n\theta}{(n+1)}$$

$$= \frac{r}{n+1} \cdot \frac{1}{\sin(n\theta)}$$

$$= \frac{r}{n+1} \cdot \frac{r^n}{a^n}$$

$$\therefore r = a \sin(n\theta)$$

$$= \frac{a^n}{(n+1) r^{n-1}} \text{ Ans.}$$

⑧ Easy $r(1+\cos\theta) = a$

$$r = \frac{a}{(1+\cos\theta)}$$

$$\rho = \left[r^2 + \left(\frac{dr}{d\theta} \right)^2 \right]^{3/2}$$

$$\frac{r}{\theta} = \frac{-a(-\sin\theta)}{(1+\cos\theta)^2} = \frac{a \sin\theta}{(1+\cos\theta)^2}$$

$$r^2 + 2 \left(\frac{dr}{d\theta} \right)^2 - r \frac{d^2 r}{d\theta^2}$$

$$\frac{d^2 r}{d\theta^2} = \frac{(1+\cos\theta)^2 a \cos\theta - a \sin^2 \theta (1+\cos\theta)}{(1+\cos\theta)^4}$$

$$= \frac{a(1+\cos\theta) \left[(1+\cos\theta) \cos\theta + 2 \sin^2 \theta \right]}{(1+\cos\theta)^4}$$

$$= \frac{a \left[(1+\cos\theta) \cos\theta + 2(1-\cos^2\theta) \right]}{(1+\cos\theta)^3}$$

$$= \frac{a(1+\cos\theta) \left[\cos\theta + 2(1-\cos\theta) \right]}{(1+\cos\theta)^3}$$

7.7-11

$$= a \frac{(\cos \theta + 2 - 2\cos \theta)}{(1 + \cos \theta)^2}$$

$$\frac{d^2 r}{d\theta^2} = a \frac{(2 - \cos \theta)}{(1 + \cos \theta)^2}$$

from ①

$$\left[r^2 + \frac{a^2 \sin^2 \theta}{(1 + \cos \theta)^4} \right]^{3/2}$$

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$$s = \frac{r^2 + 2 \frac{a^2 \sin^2 \theta}{(1 + \cos \theta)^4} - r a (2 - \cos \theta)}{(1 + \cos \theta)^2}$$

$$= \frac{[r^2 (1 + \cos \theta)^4 + a^2 \sin^2 \theta]^{3/2}}{(1 + \cos \theta)^{4 \times 3/2}}$$

$$\frac{r^2 (1 + \cos \theta)^4 + 2a^2 \sin^2 \theta - r(1 + \cos \theta)^2 a (2 - \cos \theta)}{(1 + \cos \theta)^4}$$

$$= \frac{(r^2 (1 + \cos \theta)^4 + a^2 \sin^2 \theta)^{3/2}}{(1 + \cos \theta)^6}$$

$$\frac{r^2 (1 + \cos \theta)^4 + 2a^2 \sin^2 \theta - r(1 + \cos \theta)^2 a (2 - \cos \theta)}{(1 + \cos \theta)^4}$$

∴

$$s = \frac{(r^2 (1 + \cos \theta)^4 + a^2 \sin^2 \theta)^{3/2}}{(1 + \cos \theta)^2 [r^2 (1 + \cos \theta)^4 + 2a^2 \sin^2 \theta - r(1 + \cos \theta)^2 a (2 - \cos \theta)]} \quad (11)$$

$$\text{Since } r(1 + \cos \theta) = a.$$

$$\therefore 1 + \cos \theta = \frac{a}{r}$$

$$\cos \theta = \frac{a}{r} - 1 = \frac{a - r}{r}$$

$$\cos^2 \theta = \left(\frac{a - r}{r} \right)^2$$

$$\therefore 1 - \cos^2 \theta = \sin^2 \theta.$$

$$1 - \cos^2 \theta = 1 - \left(\frac{a - r}{r} \right)^2$$

$$\therefore \sin^2 \theta = \frac{r^2 - (a^2 + r^2 - 2ar)}{r^2}$$

$$= \frac{r^2 - a^2 - r^2 + 2ar}{r^2}$$

$$\sin^2 \theta = \frac{2ar - a^2}{r^2}$$

7.7-12

from ⑪

$$I = \left(r^2 \left(\frac{a}{r} \right)^4 + a^2 \frac{(2ar - a^2)}{r^2} \right)^{3/2}$$

$$\left(\frac{a}{r} \right)^2 \left\{ r^2 \frac{a^4}{r^4} + 2a^2 \frac{(2ar - a^2)}{r^2} - r \left(\frac{a}{r} \right)^2 a \left(2 - \frac{(a-r)}{r} \right) \right\}$$

$$I = \left[\frac{a^4}{r^2} + a^2 \frac{(2ar - a^2)}{r^2} \right]^{3/2}$$

$$\frac{a^2}{r^2} \left[\frac{a^4}{r^2} + \frac{4a^3r - 2a^4}{r^2} - \frac{a^2 a (2r - a + r)}{r} \right]$$

$$I = (a^4 + 2a^3r - a^4)^{3/2}$$

$$\frac{r^3 a^2}{r^2} \left(a^4 + 4a^3r - 2a^4 - a^3 2r + a^4 - a^3 r \right)$$

$$I = \frac{(2a^3r)^{3/2}}{r^3} \cdot r^4$$

$$r^2 (4a^3r - 3a^3r)$$

$$I = \frac{2^{3/2} a^{3/2} r^{3/2}}{a^2 a^3 r} = \frac{2^{3/2} a^{3/2} r^{3/2}}{a^5 r}$$

$$= 2^{3/2} a^{-5/2} r^{1/2}$$

$$= 2^{3/2} a^{-5/2} r^{1/2}$$

$$= \frac{2\sqrt{2}}{\sqrt{a}} r^{1/2} \text{ Ans.}$$

7.7-13

Difficult

$$\textcircled{8} \quad r(1+\cos\theta) = a$$

$$\frac{dr}{d\theta}(1+\cos\theta) + r(-\sin\theta) = 0$$

$$\frac{dr}{d\theta}(1+\cos\theta) = r \sin\theta$$

$$\frac{r}{\frac{dr}{d\theta}} = \frac{r(1+\cos\theta)}{r \sin\theta}$$

$$\frac{r}{dr/d\theta} = \frac{1+\cos\theta}{\sin\theta} = \frac{2\cos^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}$$

$$\tan\phi = \cot\frac{\theta}{2}$$

$$\tan\phi = \tan\left(\frac{\pi}{2} - \frac{\theta}{2}\right)$$

$$\phi = \frac{\pi}{2} - \frac{\theta}{2}$$

We know $p = r \sin\phi$

$$\therefore p = r \sin\left(\frac{\pi}{2} - \frac{\theta}{2}\right) = r \cos\frac{\theta}{2}$$

$$\text{Squaring } p^2 = r^2 \cos^2\frac{\theta}{2} = r^2 \left(\frac{1+\cos\theta}{2}\right)$$

$$p^2 = \frac{r^2}{2} \left(\frac{a}{r}\right) = \frac{ar}{2} \quad \because r(1+\cos\theta) = a$$

$$p = \frac{\sqrt{ar}}{\sqrt{2}}$$

$$\frac{dp}{dr} = \frac{\sqrt{a}}{\sqrt{2}} \cdot \frac{1}{2\sqrt{r}} = \frac{\sqrt{a}}{2^{3/2} \sqrt{r}}$$

$$\frac{dr}{dp} = \frac{2^{3/2} r^{1/2}}{a^{1/2}}$$

707-14

$$\rho = r \frac{dr}{dp} = \frac{r \cdot r^{\frac{1}{2}} \cdot 2^{\frac{3}{2}}}{\sqrt{a}} = \frac{r^{\frac{3}{2}} \cdot 2^{\frac{3}{2}}}{\sqrt{a}} \quad \text{Ans}$$

Q9. $x^2y = a(x^2 + y^2)$ Implicit

$$\rho = (f_x^2 + f_y^2)^{\frac{3}{2}}$$

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$$f_y^2(f_{xx}) + f_x^2(f_{yy}) - 2f_x f_y f_{xy}$$

Here $f(x, y) = a(x^2 + y^2) - x^2y$

$$f_x = a(2x) - 2xy$$

$$f_y = a(2y) - x^2$$

$$f_{xy} = -2x$$

$$f_{xx} = a(2) = 2a$$

$$f_{yy} = a(2)$$

At $(2a, 2a)$

$$f_x = a[2(2a)] - 2(2a)(2a) = 4a^2 - 8a^2 = -4a^2$$

$$f_y = a[2(2a)] - (2a)^2 = 4a^2 - 4a^2 = 0$$

$$f_{xy} = -2(2a) = -4a$$

$$f_{xx} = 2a - 2(2a) = 2a - 4a = -2a$$

$$f_{yy} = 2a$$

Therefore $\rho = \left[(-4a^2)^2 + 0 \right]^{\frac{3}{2}} = \frac{(16a^4)^{\frac{3}{2}}}{0 + (-4a^2)^2(2a) - 2(-4a^2)(0)(-4a)}$

$$= \frac{(4a^2)^{\frac{6}{2}}}{32a^5} = \frac{64a^6}{32a^5} = 2a \quad \text{Ans}$$

7.7-15

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10) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

We use parametric eqs. $x = a \cos \theta$
 $y = b \sin \theta$

$$S = \left\{ \left(\frac{dx}{d\theta} \right)^2 + \left(\frac{dy}{d\theta} \right)^2 \right\}^{3/2}$$

$$\left(\frac{dx}{d\theta} \right) \left(\frac{d^2y}{d\theta^2} \right) - \left(\frac{dy}{d\theta} \right) \left(\frac{d^2x}{d\theta^2} \right)$$

$$S = \left\{ (-a \sin \theta)^2 + (b \cos \theta)^2 \right\}^{3/2}$$

$$(-a \sin \theta)(-b \sin \theta) - (b \cos \theta)(-a \cos \theta)$$

$$S = \frac{(a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{3/2}}{ab \sin^2 \theta + ab \cos^2 \theta}$$

$$S = \frac{(a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{3/2}}{ab(1)}$$

Eq of Tangent.

$$y - y_1 = m(x - x_1)$$

$$y - (b \sin \theta) = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$$

$$bx \cos \theta + ay \sin \theta = ab \cos^2 \theta + ab \sin^2 \theta$$

$$bx \cos \theta + ay \sin \theta = ab(\cos^2 \theta + \sin^2 \theta)$$

$$\frac{dx}{d\theta} = a(-\sin \theta)$$

$$\frac{d^2x}{d\theta^2} = -a \cos \theta$$

$$\frac{dy}{d\theta} = b \cos \theta$$

$$\frac{d^2y}{d\theta^2} = -b \sin \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$\frac{dy}{dx} = \frac{b \cos \theta}{a(-\sin \theta)}$$

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7.7-16

$$bx \cos \theta + ay \sin \theta = ab$$

$$bx \cos \theta + ay \sin \theta - ab = 0$$

origin (0,0)

$$p = \frac{|bx \cos \theta + ay \sin \theta - ab|}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$= \frac{|b(0) \cos \theta + a(0) \sin \theta - ab|}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$p = \frac{|ab|}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \Rightarrow p^2 = \frac{a^2 b^2}{b^2 \cos^2 \theta + a^2 \sin^2 \theta}$$

$$a^2 \sin^2 \theta + b^2 \cos^2 \theta = \frac{a^2 b^2}{p^2} \quad \text{Put in (1)}$$

$$p = \frac{\left(\frac{a^2 b^2}{p^2}\right)^{3/2}}{ab} = \frac{\frac{a^3 b^3}{p^3}}{ab} = \left[\frac{a^2 b}{p}\right]$$

If CQ is semidiameter conjugate to CP, then coordinates of Q are $(-a \sin \theta, b \cos \theta)$. C(0,0) is centre of ellipse.

$$|CQ| = \sqrt{(-a \sin \theta - 0)^2 + (b \cos \theta - 0)^2} = \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$|CQ|^2 = a^2 \sin^2 \theta + b^2 \cos^2 \theta \quad \text{Put in (1)}$$

$$p = \frac{(a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{3/2}}{ab} = \frac{[|CQ|^2]^{3/2}}{ab} = \frac{|CQ|^3}{ab}$$

7-7-18

End Method

① $y^2 = 4ax$

$$2y \frac{dy}{dx} = 4a \Rightarrow \frac{dy}{dx} = \frac{2a}{y} = \frac{2a}{y}$$

$$\frac{d^2y}{dx^2} = -\frac{2a}{y^2} \left(\frac{dy}{dx} \right) = -\frac{2a}{y^2} \left(\frac{2a}{y} \right) = -\frac{4a^2}{y^3}$$

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}{\frac{d^2y}{dx^2}}$$

$$= \frac{\left[1 + \left(\frac{2a}{y} \right)^2 \right]^{3/2}}{-\frac{4a^2}{y^3}} = \frac{\left[\frac{y^2 + 4a^2}{y^2} \right]^{3/2}}{-\frac{4a^2}{y^3}}$$

$$= \frac{(y^2 + 4a^2)^{3/2}}{y^3} \cdot \left(\frac{y^3}{-4a^2} \right) = \frac{(y^2 + 4a^2)^{3/2}}{-4a^2} = \frac{(4ax + 4a^2)^{3/2}}{-4a^2} = \frac{(4a(x+a))^{3/2}}{-4a^2}$$

$$\rho = \frac{(2^2)^{3/2} \cdot a^{3/2} (x+a)^{3/2}}{-4a^2} = \frac{2^3 a^{3/2} (x+a)^{3/2}}{-4a^2} = \frac{8(x+a)^{3/2}}{-4a^{2-3/2}} = \frac{8(x+a)^{3/2}}{-4a^{1/2}}$$

$$\rho = \frac{2(x+a)^{3/2}}{-\sqrt{a}}$$

$$\rho^2 = \frac{4(x+a)^3}{a} \quad \text{--- ① ---}$$

$$|SP| = \sqrt{(x-a)^2 + (y-0)^2} = \sqrt{x^2 + a^2 - 2ax + y^2}$$

$$|SP| = \sqrt{x^2 + a^2 - 2ax + 4ax} = \sqrt{x^2 + a^2 + 2ax} = \sqrt{(x+a)^2} = x+a$$

from ① $\rho^2 = \frac{4|SP|^3}{a}$ $\rho \propto |SP|^3$ $\frac{4}{a}$ is const.

$$f(x,y) = 4ax - y^2$$

$$f_x = 4a, f_y = -2y$$

$$f_{xx} = 0, f_{yy} = -2, f_{xy} = 0$$

$$\rho = (f_{xx}^2 + f_{yy}^2)^{3/2}$$

$$f_y f_{xx} - 2f_x f_y f_{xy} + f_x^2 f_{yy}$$

$$= (16a^2 + 4y^2)^{3/2}$$

$$4y^2(0) - 2(4a)(-2y)(0) + (4a)^2(-2)$$

$$\rho = \frac{(16a^2 + 4(4ax))^{3/2}}{-32a^2}$$

$$\rho = \frac{(16a^2)^{3/2} (a+x)^{3/2}}{-32a^2} = \frac{(4)^{3/2} a^{3/2} (a+x)^{3/2}}{-32a^2}$$

$$\rho = \frac{-2(a+x)^{3/2}}{\sqrt{a}}$$

$$\because y^2 = 4ax$$

we may use here $\rho(x,y)$ but then $\rho = \frac{4}{a}(x+a)^{3/2}$

$$\rho(x,y)$$

$$f = (a,0) \text{ focus of parabola}$$

$$\because y^2 = 4ax$$

7.7-18

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Imp.

⑫ $r = a(1 + \cos \theta)$

$$\frac{r}{a} = 1 + \cos \theta$$

$$\rho = r \frac{dr}{dp}$$

$$\frac{dr}{d\theta} = -a \sin \theta$$

$$\rho = r \sin \psi$$

$$\frac{d^2 r}{d\theta^2} = -a \cos \theta$$

$$\tan \psi = \frac{r}{\frac{dr}{d\theta}}$$

$$\tan \psi = \frac{r}{\frac{dr}{d\theta}} = \frac{a(1 + \cos \theta)}{-a \sin \theta}$$

$$= \frac{2 \cos^2 \frac{\theta}{2}}{-2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$$

$$\tan \psi = -\cot \frac{\theta}{2}$$

$$\tan \psi = \tan \left(\frac{\pi}{2} + \frac{\theta}{2} \right)$$

$$\psi = \frac{\pi}{2} + \frac{\theta}{2}$$

We know $\rho = r \sin \psi$

$$\rho = r \sin \left(\frac{\pi}{2} + \frac{\theta}{2} \right)$$

$$\rho = r \cos \frac{\theta}{2}$$

$$\rho^2 = r^2 \cos^2 \frac{\theta}{2}$$

$$\rho^2 = \frac{r^2}{2} (2 \cos^2 \frac{\theta}{2}) = \frac{r^2}{2} (1 + \cos \theta)$$

$$\rho^2 = \frac{r^2}{2} \left(\frac{r}{a} \right) = \frac{r^3}{2a}$$

$$\rho = \frac{r^{3/2}}{\sqrt{2} a}$$

$$\frac{d\rho}{dr} = \frac{1}{\sqrt{2} a} \left(\frac{3}{2} r^{\frac{3}{2}-1} \right) = \frac{3}{2\sqrt{2} a} r^{\frac{1}{2}}$$

[Ind Method

$$\frac{dr}{d\theta} = -a \sin \theta \quad \text{Easy}$$

$$\frac{d^2 r}{d\theta^2} = -a \cos \theta$$

$$\rho = \left[r^2 + \left(\frac{dr}{d\theta} \right)^2 \right]^{3/2}$$

$$= \left[a^2 (1 + \cos \theta)^2 + a^2 \sin^2 \theta \right]^{3/2}$$

$$= a^2 (1 + \cos \theta)^2 + 2a^2 \cos \theta - a^2 (1 + \cos \theta)(-a \cos \theta)$$

$$= \left[a^2 (1 + \cos^2 \theta + 2 \cos \theta + \sin^2 \theta) \right]^{3/2}$$

$$= a^2 (1 + \cos^2 \theta + 2 \cos \theta + 2 \sin^2 \theta + \cos \theta + \cos^2 \theta)$$

$$= \frac{3}{2} a^2 (1 + 2 \cos \theta)^{3/2}$$

$$= \frac{3}{2} a^2 (1 + 2 \cos^2 \theta + 2 \sin^2 \theta + 3 \cos \theta)$$

$$= \frac{3}{2} a^2 (2 + 2 \cos \theta)^{3/2}$$

$$(1 + 2 + 3 \cos \theta)$$

$$= \frac{3}{2} a^2 (1 + \cos \theta)^{3/2}$$

$$3 (1 + \cos \theta)$$

7-7-19

$\frac{dr}{dp} = \frac{2\sqrt{2}a}{3\sqrt{r}}$	$\rho = \frac{a2\sqrt{2}}{3} (1+\cos\theta)^{\frac{3}{2}-1}$
$\rho = \frac{r \frac{dr}{dp}}{\frac{dr}{dp}} = \frac{r \frac{2\sqrt{2}a}{3\sqrt{r}}}{\frac{dr}{dp}}$	$\rho = \frac{a2\sqrt{2}}{3} (1+\cos\theta)^{\frac{1}{2}}$ squaring
$\rho = \frac{r^{\frac{1}{2}} 2\sqrt{2}a}{3}$	$\rho^2 = a^2 8 (1+\cos\theta)$ ← (Remember) for Q17, 18
ing, $\rho = \frac{\pi 4(2)(a)}{9} = \frac{8a\pi}{9}$	$\rho^2 = \frac{a^2 8}{9} \frac{\pi}{a}$
$\frac{\rho^2}{r} = \frac{8a}{9} = \text{Constant}$	$\frac{\rho^2}{r} = \frac{8a}{9} = \text{Const.}$

Q17 Let PA be a chord through pole 'O'

then $P=(r, \alpha)$ & $Q=(r', \alpha+\pi)$

$$r = a(1+\cos\theta)$$

As I solved above in Exam Solve completely

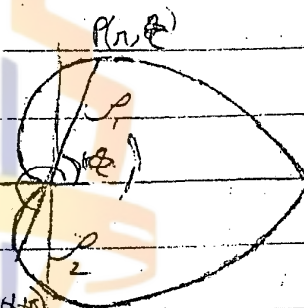
$$\rho^2 = \frac{8a^2}{9} (1+\cos\theta)$$

$$\rho_1^2 = \frac{8a^2}{9} (1+\cos\theta) \quad \text{at } P(r, \alpha)$$

$$\rho_2^2 = \frac{8a^2}{9} (1+\cos(\pi+\theta)) \quad \text{at } Q(r', \alpha+\pi)$$

$$\rho_2^2 = \frac{8a^2}{9} (1-\cos\theta)$$

$$\begin{aligned} \rho_1^2 + \rho_2^2 &= \frac{8a^2}{9} (1+\cos\theta) + \frac{8a^2}{9} (1-\cos\theta) \\ &= \frac{8a^2}{9} [1+\cos\theta + 1-\cos\theta] = \frac{8a^2}{9} (2) = \frac{16a^2}{9} \end{aligned}$$



7.7-20

Q18 $r = a(1 + \cos \theta)$ ①

$\therefore \rho^2 = \frac{8a^2}{9}(1 + \cos \theta)$ (As I solved in Q12)
I can solve completely

$\frac{dr}{d\theta} = -a \sin \theta$

$\frac{d^2r}{d\theta^2} = -a \cos \theta$

$\tan \psi = \frac{r}{dr/d\theta} = \frac{a(1 + \cos \theta)}{-a \sin \theta} = \frac{2 \cos^2 \theta/2}{-2 \sin \theta/2 \cos \theta/2} = \frac{\cos \theta/2}{-\sin \theta/2}$

$\tan \psi = -\cot \frac{\theta}{2} = \tan \left(\frac{\pi}{2} + \frac{\theta}{2} \right) \Rightarrow \psi = \frac{\pi}{2} + \frac{\theta}{2}$

Let α be the angle which tangent makes then $\alpha = \theta + \psi$

$\therefore \alpha = \theta + \frac{\pi}{2} + \frac{\theta}{2} = \frac{3\theta}{2} + \frac{\pi}{2}$ ②

Now tangent is // to initial line when $\alpha = 0, \pi$

when $\alpha = 0$ use ② $0 = \frac{3\theta}{2} + \frac{\pi}{2} \Rightarrow \frac{3\theta}{2} = -\frac{\pi}{2} \Rightarrow \theta = -\frac{\pi}{3}$

when $\alpha = \pi$ use ② $\pi = \frac{3\theta}{2} + \frac{\pi}{2} \Rightarrow \frac{3\theta}{2} = \pi - \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{3}$

from ① At $\theta = -\frac{\pi}{3}$ $\rho^2 = \frac{8a^2}{9} \left(1 + \cos \left(-\frac{\pi}{3} \right) \right)$

$= \frac{8a^2}{9} \left(1 + \cos \frac{\pi}{3} \right) = \frac{8a^2}{9} \left(1 + \frac{1}{2} \right) = \frac{8a^2}{9} \left(\frac{3}{2} \right) = \frac{4a^2}{3}$

$\rho = \frac{4a^2}{3} \Rightarrow \rho = \frac{2a}{\sqrt{3}}$

At $\theta = \frac{\pi}{3}$ $\rho^2 = \frac{8a^2}{9} \left(1 + \cos \frac{\pi}{3} \right) = \frac{8a^2}{9} \left(1 + \frac{1}{2} \right) = \frac{8a^2}{9} \left(\frac{3}{2} \right) = \frac{4a^2}{3}$

$\therefore \rho = \frac{2a}{\sqrt{3}}$

7-7-21

Q. 13. $\rho = r \frac{dr}{dp}$ To Prove. Don't use $\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{1/2}$ it becomes very difficult.

Proof. We know $\rho = r \sin \psi$

$$\frac{dp}{dr} = 1 \cdot \sin \psi + r \cos \psi \frac{d\psi}{dr} \quad \text{--- (1)}$$

Since $\tan \psi = r \frac{d\theta}{dr}$ $\therefore \sin \psi = \frac{r d\theta}{ds}$

$\Rightarrow \tan \psi = r \frac{d\theta}{dr/ds}$ $\& \cos \psi = \frac{dr}{ds}$

Becomes $\frac{dp}{dr} = 1 \cdot \frac{r d\theta}{ds} + r \frac{dr}{ds} \frac{d\psi}{dr}$

$$\frac{dp}{dr} = r \frac{d\theta}{ds} + r \frac{d\psi}{ds}$$

$$= r \frac{d(\theta + \psi)}{ds}$$

$$= r \frac{d\alpha}{ds}$$

$$\therefore \alpha = \theta + \psi$$

$$\frac{dp}{dr} = r \left(\frac{1}{\rho}\right)$$

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$$\frac{d\alpha}{ds} = K = \frac{1}{\rho}$$

$$\rho = r \frac{dr}{dp} \quad \text{Proved.}$$

7.7-22

(14) $p^2 = ar$

$$2p \frac{dp}{dr} = a$$

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$$\frac{dp}{dr} = \frac{a}{2p} \Rightarrow \frac{dr}{dp} = \frac{2p}{a}$$

$$r = r \left(\frac{dr}{dp} \right) = r \left(\frac{2p}{a} \right) = \frac{p^2}{a} \left(\frac{2p}{a} \right) \quad \therefore r = \frac{p^2}{a}$$

$$r = \frac{2p^3}{a^2} \text{ Ans.}$$

(15) $\frac{1}{p^2} = \frac{A}{r^2} + B$

$$\frac{-2}{p^3} \frac{dp}{dr} = -\frac{2A}{r^3} + 0$$

$$\frac{dp}{dr} = -\frac{2A}{r^3} \left(\frac{p^3}{-2} \right) = \frac{Ap^3}{r^3}$$

$$\frac{dr}{dp} = \frac{r^3}{Ap^3}$$

$$r = \frac{r dr}{dp} = \frac{r \cdot r^3}{Ap^3} = \frac{r^4}{Ap^3} \text{ Ans. in the Book}$$

$$r = \frac{(Ap^2)^2}{(1-p^2B)^2} \cdot \frac{1}{Ap^3} = \frac{A^2 p^4}{Ap^3 (1-p^2B)^2}$$

$$\frac{1}{p^2} - B = \frac{A}{r^2}$$

$$\frac{1-p^2B}{p^2} = \frac{A}{r^2}$$

$$r = \frac{Ap}{(1-p^2B)^2} \text{ Ans (I think this is the right one)}$$

$$r^2 = \frac{Ap^2}{1-p^2B}$$

7-7-23

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$$p^2(a^2+b^2-r^2) = a^2b^2$$

$$2p \frac{dp}{dr}(a^2+b^2-r^2) + p^2(0+0-2r) = 0$$

$$2p \frac{dp}{dr}(a^2+b^2-r^2) - 2p^2r = 0$$

$$2p \frac{dp}{dr}(a^2+b^2-r^2) = 2p^2r$$

$$\frac{dp}{dr}(a^2+b^2-r^2) = \frac{2p^2r}{2p} = pr$$

$$\frac{dp}{dr} = \frac{pr}{(a^2+b^2-r^2)} \Rightarrow \frac{dr}{dp} = \frac{(a^2+b^2-r^2)}{pr}$$

$$p = r \left(\frac{dr}{dp} \right) = r \cdot \frac{(a^2+b^2-r^2)}{pr}$$

$$p = \frac{1}{p} (a^2+b^2-r^2) = \frac{1}{p} \left(\frac{a^2b^2}{p^2} \right)$$

$$\because p^2(a^2+b^2-r^2) = a^2b^2$$

$$p = \frac{a^2b^2}{p^3} \text{ Ans.}$$

$$\text{Q19 } y = ax^2 + bx + c$$

$$\frac{dy}{dx} = 2ax + b$$

$$\frac{d^2y}{dx^2} = 2a$$

$$p = \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2} = \frac{[1 + (2ax+b)^2]^{3/2}}{2a}$$

$$y = ax^2 + bx + c$$

$$\frac{y}{a} = x^2 + \frac{b}{a}x + \frac{c}{a}$$

$$\frac{y}{a} = x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} + \frac{c}{a}$$

$$\frac{y}{a} - \frac{c}{a} + \frac{b^2}{4a^2} = \left(x + \frac{b}{2a} \right)^2$$

$$\left(\frac{4ay - 4ac + b^2}{4a^2} \right) = \left(x + \frac{b}{2a} \right)^2$$

$$\sqrt{\frac{4ay - 4ac + b^2}{4a^2}} = \left(x + \frac{b}{2a} \right) \Leftrightarrow \frac{1}{a} \left(y - \frac{4ac - b^2}{4a} \right) = \left(x + \frac{b}{2a} \right) \Leftrightarrow \frac{y}{a} - \frac{(4ac - b^2)}{4a^2} = \left(x + \frac{b}{2a} \right)^2$$

7.7-24

For extreme values of P , Put $\frac{dP}{dx} = 0$

$$\frac{dP}{dx} = \frac{3}{2} \frac{(1+(2ax+b)^2)^{\frac{1}{2}} \cdot 2(2ax+b)2a}{2a} = 3(2ax+b)(1+(2ax+b)^2)^{\frac{1}{2}}$$

$$0 = 3(2ax+b)(1+(2ax+b)^2)^{\frac{1}{2}}$$

either $(2ax+b) = 0$ or $(1+(2ax+b)^2)^{\frac{1}{2}} = 0$

$$\Rightarrow 2ax+b = 0$$

$$(1+(2ax+b)^2)^{\frac{1}{2}} = 0$$

$$\boxed{x = -\frac{b}{2a}}$$

Squaring $1+(2ax+b)^2 = 0$

$$(2ax+b)^2 = -1$$

$$2ax+b = \pm i$$

Discard being Imaginary.

$$\frac{d^2P}{dx^2} = 3(2a)(1+(2ax+b)^2)^{\frac{1}{2}} + 3(2ax+b) \cdot \frac{1}{2} \cdot 2(2ax+b)2a$$

$$\frac{d^2P}{dx^2} = \frac{6a(1+(2ax+b)^2) + 6a(2ax+b)^2}{1+(2ax+b)^2}$$

At $x = -\frac{b}{2a}$ i.e. $2ax+b = 0$

$$\frac{d^2P}{dx^2} = \frac{6a(1+0) + 6a(0)}{1+0} = 6a > 0 \text{ Hence } P \text{ is minimum at } x = -\frac{b}{2a}$$

For Pb of Verten $y = ax^2 + bx + c \Rightarrow y = a\left(-\frac{b}{2a}\right)^2 + b\left(-\frac{b}{2a}\right) + c$

$$\Rightarrow y = \frac{a b^2}{4a^2} - \frac{b^2}{2a} + c \Rightarrow y = \frac{b^2}{4a} - \frac{b^2}{2a} + c \Rightarrow y = \frac{b^2 - 2b^2 + 4ac}{4a}$$

$$y = \frac{-b^2 + 4ac}{4a}$$

So Verten of Parabola is $\left(-\frac{b}{2a}, \frac{-b^2 + 4ac}{4a}\right)$

7.7-25

⑩ $y = \ln x$

$$\frac{dy}{dx} = \frac{1}{x}$$

$$\frac{d^2y}{dx^2} = -\frac{1}{x^2}$$

Curvature at any pt (x, y) of the curve is

$$K = \frac{|d^2y/dx^2|}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}$$

$$K = \frac{\left|-\frac{1}{x^2}\right|}{\left(1 + \frac{1}{x^2}\right)^{3/2}} = \frac{\frac{1}{x^2}}{\frac{(x^2+1)^{3/2}}{x^3}} = \frac{1 \cdot x^3}{x^2(x^2+1)^{3/2}}$$

$$K = \frac{x}{(x^2+1)^{3/2}}$$

For extrem values Put $\frac{dK}{dx} = 0$

$$\frac{dK}{dx} = (x^2+1)^{3/2} (1) - (x) \frac{3}{2} (x^2+1)^{1/2} \cdot 2x = \frac{(x^2+1)^{3/2} - 3x^2(x^2+1)^{1/2}}{(x^2+1)^3}$$

$$\frac{dK}{dx} = (x^2+1)^{1/2} \frac{(x^2+1 - 3x^2)}{(x^2+1)^3} = \frac{1-2x^2}{(x^2+1)^{5/2}}$$

$$0 = \frac{1-2x^2}{(x^2+1)^{5/2}} \Rightarrow 1-2x^2 = 0 \Rightarrow x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}}$$

$$\frac{d^2K}{dx^2} = (x^2+1)^{5/2} (-4x) - (1-2x^2) \frac{5}{2} (x^2+1)^{3/2} \cdot 2x$$

7.7-26

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$$\frac{-(x^2+1)^{3/2} \left[(x^2+1)(-4x) - 5x(1-2x^2) \right]}{(x^2+1)^5}$$

$$\frac{d^2K}{dx^2} = \frac{\{(x^2+1)(-4x) - 5x(1-2x^2)\}}{(x^2+1)^{7/2}}$$

At $x = \frac{1}{\sqrt{2}}$

$$\frac{d^2K}{dx^2} = \frac{\left\{ \left(\frac{1}{2} + 1 \right) \left(-4 \frac{1}{\sqrt{2}} \right) - 5 \frac{1}{\sqrt{2}} \left(1 - 2 \frac{1}{2} \right) \right\}}{\left(\frac{1}{2} + 1 \right)^{7/2}}$$

$$= \frac{\left(\frac{3}{2} \right) \left(-2\sqrt{2} \right) - 0}{\left(\frac{3}{2} \right)^{7/2}}$$

$$= \frac{-3\sqrt{2}}{\left(\frac{3}{2} \right)^{7/2}}$$

Hence P is

Max at $x = \frac{1}{\sqrt{2}}$

For $x = \frac{1}{\sqrt{2}}$

$$y = \ln x$$

$$= \ln \left(\frac{1}{\sqrt{2}} \right) = \ln (2)^{-1/2}$$

$$= (-1) \ln 2^{1/2}$$

$$y = (-1) \ln \sqrt{2}$$

Hence required pt is $\left(\frac{1}{\sqrt{2}}, -\ln \sqrt{2} \right)$

x

7.7 - 27

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$$\text{Prove } \rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}$$

Radius of Curvature
for Cartesian Curve.

Let $y = f(x)$ be the eq of the curve and ψ be the inclination of the tangent to the curve at $x = a$.

$$\text{then } \tan \psi = \frac{dy}{dx}$$

Diff w.r.t. 's'

$$\text{Sec}^2 \psi \frac{d\psi}{ds} = \frac{d^2y}{dx^2} \frac{dx}{ds} \quad \left/ \frac{d}{ds} \left(\frac{dy}{dx} \right) \right.$$

$$= \frac{d}{dx} \left(\frac{dy}{dx} \right) \frac{dx}{ds}$$

$$\therefore \frac{d\psi}{ds} = \frac{\frac{d^2y}{dx^2}}{\text{Sec}^2 \psi \frac{ds}{dx}}$$

$$ds^2 = (dx)^2 + (dy)^2$$

$$\frac{ds^2}{dx^2} = 1 + \left(\frac{dy}{dx}\right)^2$$

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\left| \frac{d\psi}{ds} \right| = \frac{\left| \frac{d^2y}{dx^2} \right|}{(1 + \tan^2 \psi) \sqrt{1 + \left(\frac{dy}{dx}\right)^2}}$$

$$K = \frac{\left| \frac{d^2y}{dx^2} \right|}{\left[1 + \left(\frac{dy}{dx}\right)^2\right] \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{1}{2}}} = \frac{\left| \frac{d^2y}{dx^2} \right|}{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}$$

$$\rho = \frac{1}{K} = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left| \frac{d^2y}{dx^2} \right|}$$

Radius of Curvature

7.7-28

Prove that $\rho = \frac{[(f')^2 + (g')^2]^{3/2}}{|f'g'' - g'f''|}$ Radius of Curvature for Parametric Eqs

Proof

Let $x = f(t)$ $y = g(t)$ be the parametric Eqs

$$\frac{dx}{dt} = f'(t) \quad \frac{dy}{dt} = g'(t)$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = g'(t) \cdot \frac{1}{f'(t)} = \frac{g'}{f'}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{g'}{f'} \right) = \frac{d}{dt} \left(\frac{g'}{f'} \right) \frac{dt}{dx}$$

$$y'' = \frac{d^2y}{dx^2} = \left(\frac{f'g'' - g'f''}{(f')^2} \right) \frac{1}{f'} = \frac{f'g'' - g'f''}{(f')^3}$$

$$K = \frac{|y''|}{[1 + (y')^2]^{3/2}} = \frac{|f'g'' - g'f''|}{(f')^3 \{1 + (g'/f')^2\}^{3/2}}$$

$$K = \frac{|f'g'' - g'f''|}{(f')^3 \{(f')^2 + (g')^2\}^{3/2}} = \frac{|f'g'' - g'f''|}{(f')^3 \{(f')^2 + (g')^2\}^{3/2}}$$

$$= \frac{f'g'' - g'f''}{\{(f')^2 + (g')^2\}^{3/2}}$$

$$\rho = \frac{1}{K}$$

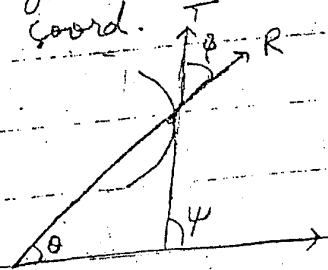
$$\therefore \rho = \frac{[(f')^2 + (g')^2]^{3/2}}{|f'g'' - g'f''|}$$

proved

7-7-29

Prove that $\rho = \frac{[r^2 + (r')^2]^{3/2}}{r^2 + 2(r')^2 - r r''}$ Radius of Curvature for curves in polar coord.

Proof Let $r = f(\theta)$ be the curve and ψ be the angle of the tangent at $P(r, \theta)$



From fig $\psi = \theta + \phi$

$$\begin{aligned} \frac{d\psi}{ds} &= \frac{d\theta}{ds} + \frac{d\phi}{ds} \\ &= \frac{d\theta}{ds} + \frac{d\phi}{d\theta} \frac{d\theta}{ds} = \frac{d\theta}{ds} \left[1 + \frac{d\phi}{d\theta} \right] \quad \text{--- ①} \end{aligned}$$

Now find $\frac{d\phi}{d\theta}$

$$\tan \phi = \frac{r}{dr/d\theta}$$

Diff w.r.t θ

$$\sec^2 \phi \frac{d\phi}{d\theta} = \frac{\left(\frac{dr}{d\theta}\right) \left(\frac{dr}{d\theta}\right) - r \frac{d^2 r}{d\theta^2}}{\left(\frac{dr}{d\theta}\right)^2}$$

$$\frac{d\phi}{d\theta} = \frac{\left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2 r}{d\theta^2}}{\sec^2 \phi \left(\frac{dr}{d\theta}\right)^2} = \frac{\left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2 r}{d\theta^2}}{(1 + \tan^2 \phi) \left(\frac{dr}{d\theta}\right)^2}$$

$$\frac{d\phi}{d\theta} = \frac{\left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2 r}{d\theta^2}}{\left[1 + r^2 \left(\frac{d\theta}{dr}\right)^2\right] \left(\frac{dr}{d\theta}\right)^2} \quad \text{Put in ①}$$

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$$\frac{dy}{ds} = \frac{d\theta}{ds} \left[\frac{1 + \left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2r}{d\theta^2}}{\left(\frac{dr}{d\theta}\right)^2 + r^2} \right]$$

$$K = \frac{d\theta}{ds} \left(\left(\frac{dr}{d\theta}\right)^2 + r^2 + \left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2r}{d\theta^2} \right)$$

$$\left(\frac{dr}{d\theta}\right)^2 + r^2$$

$$K = \frac{2\left(\frac{dr}{d\theta}\right)^2 + r^2 - r \frac{d^2r}{d\theta^2}}{\left(\left(\frac{dr}{d\theta}\right)^2 + r^2\right) \left(\frac{ds}{d\theta}\right)}$$

$$\therefore K = \frac{2(r')^2 + r^2 - r r''}{\left[(r')^2 + r^2\right] \left[\frac{1}{\sqrt{r^2 + (r')^2}}\right]}$$

$$K = \frac{r^2 + 2(r')^2 - r r''}{\left[r^2 + (r')^2\right]^{3/2}}$$

$$\mathcal{P} = \frac{1}{K} = \frac{\left[r^2 + (r')^2\right]^{3/2}}{r^2 + 2(r')^2 - r r''}$$